

ENGINEERING DRAWING EDR 101 ISOMETRIC VIEW AND PROJECTION

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COURSE CONTENT

Lecture	Topics
01	Introduction to Engineering Drawing
02	Lettering, Numbering, Dimensioning, Drawing Scales
03	Plane Geometry: Pentagon, Hexagon, Octagon, Ellipse, Parabola, Hyperbola
04	Solid Object: Polyhedra, Prism, Pyramid, Cone, Cylinder, Sphere, Frustum, Truncated
05	Projection: Orthographic, Isometric, Oblique
06	Drawing isometric projections of solids; Drawing isometric views of objects
MID TERM – I	
07	Drawing orthographic projections of solids and different objects
08	Introduction to AutoCAD: Interface, Basic command, Layout, Creating different types of drawing, Dimension & Annotation, Printing & Publishing
09	Course wrap-up and evaluation
FINAL EXAMINATION	

ISOMETRIC PROJECTION

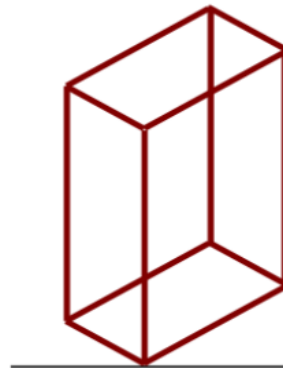
- “Iso” means “equal” and “metric” means “measure”. An isometric projection is one type of pictorial projection in which the three dimensions of a solid are not only shown in one view, but also their dimensions can be scaled from this drawing.
- Isometric projection is drawn by establishing a vertical axis and two 30-degree angled axes to create a 3D view on a 2D surface, ensuring all lines are parallel.
- In a true "isometric projection," the lines are foreshortened to approximately 81.6% (or 0.82) of their true length to simulate how they appear when tilted.

ISOMETRIC PROJECTION VS ISOMETRIC VIEW

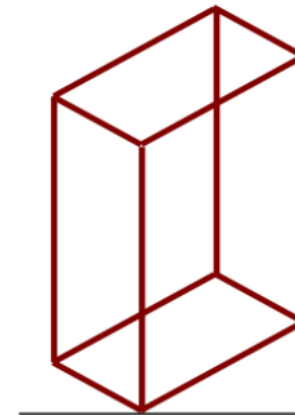
Isometric Projection	Isometric View
Drawn to isometric scale	Drawn to actual scale
When lines are drawn parallel to isometric axes, the lengths are foreshortened to 0.816 or 0.82 times the actual lengths	When lines are drawn parallel to isometric axes, the true lengths are laid off.



ORTHOGRAPHIC
PROJECTION

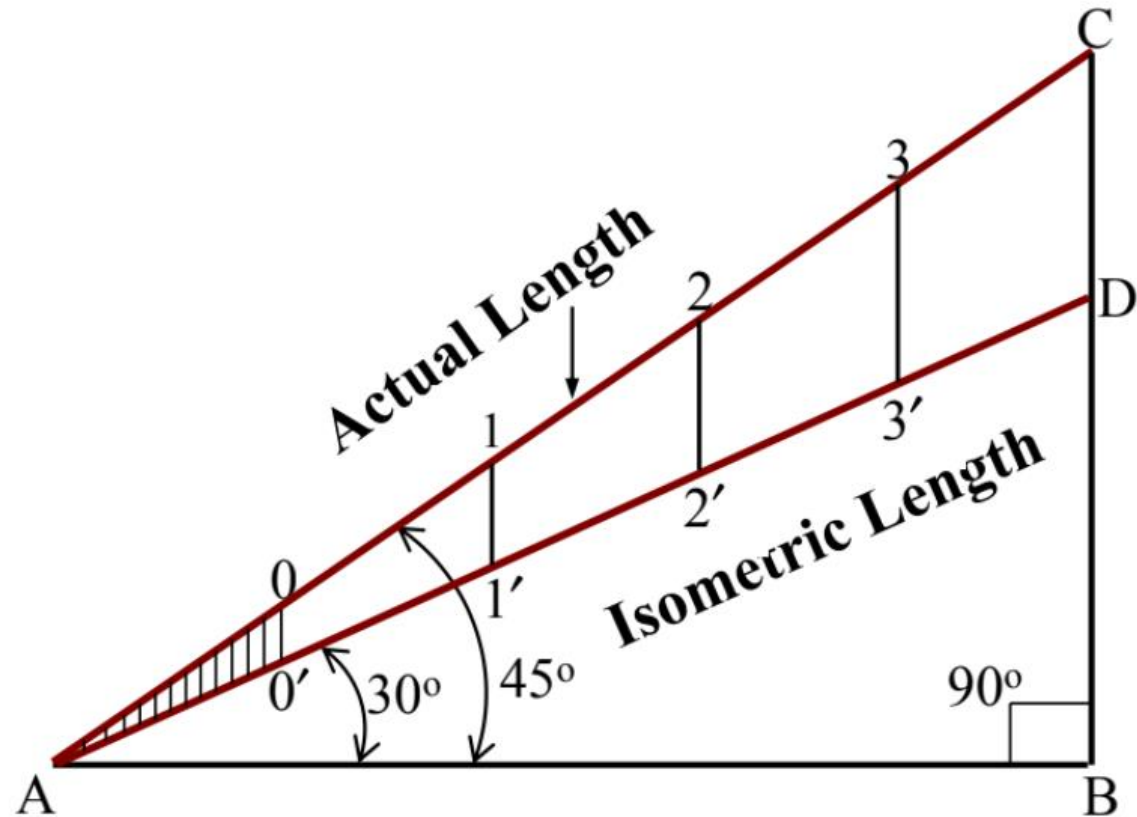


ISOMETRIC
PROJECTION



ISOMETRIC
VIEW

ISOMETRIC SCALE



$$\frac{AB}{AC} = \cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\frac{AB}{AD} = \cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\therefore \frac{AB}{AC} \div \frac{AB}{AD} = \frac{1}{\sqrt{2}} \div \frac{\sqrt{3}}{2}$$

$$\frac{AB}{AC} \times \frac{AD}{AB} = \frac{1}{\sqrt{2}} \times \frac{2}{\sqrt{3}}$$

$$\frac{AD}{AC} = \sqrt{\frac{2}{3}} = 0.81$$

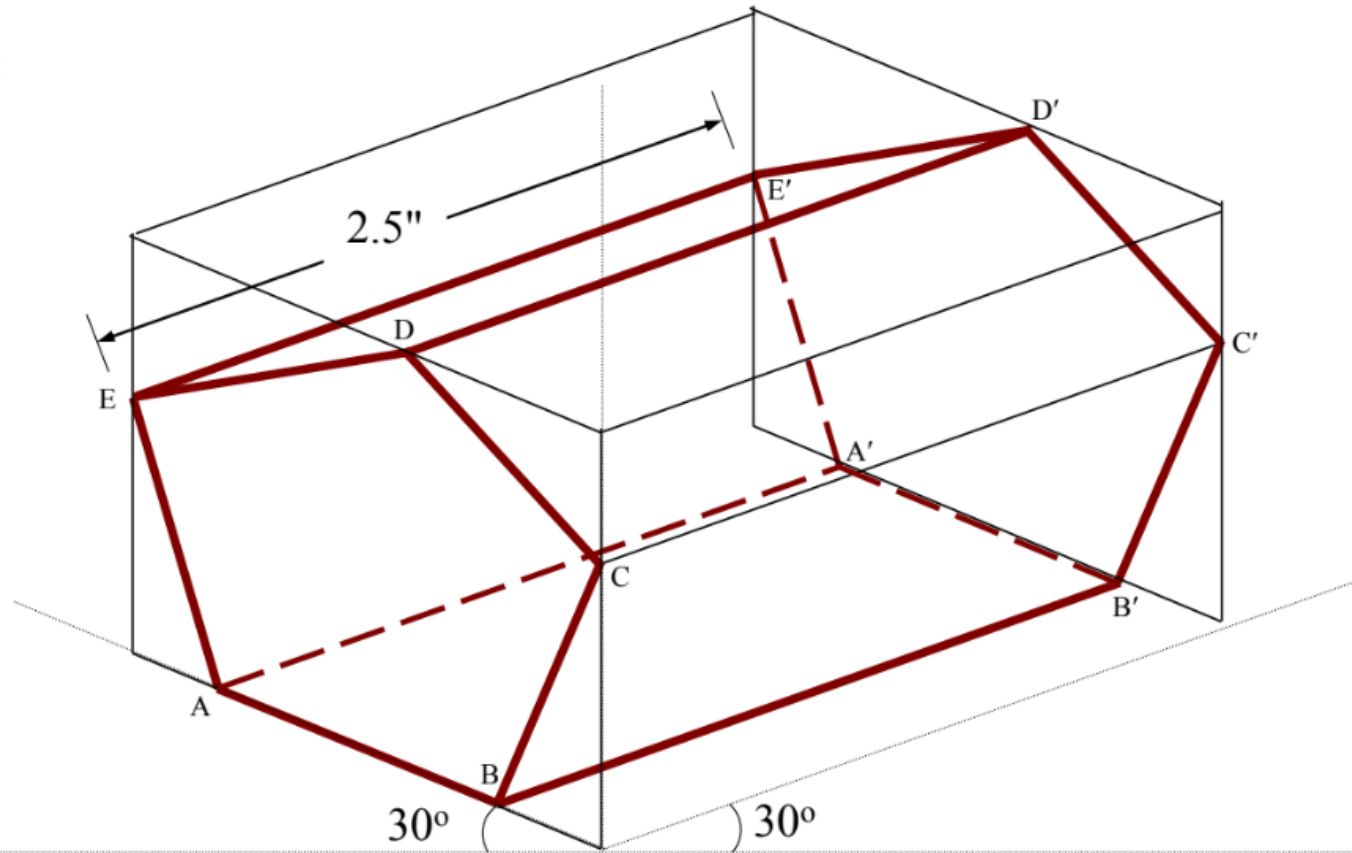
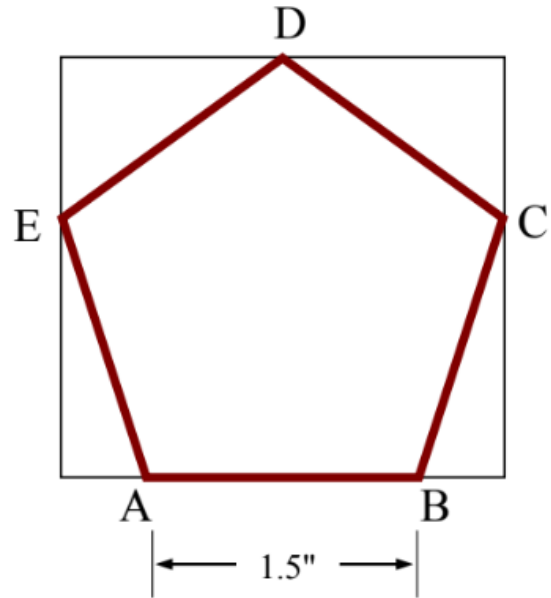
$$\Rightarrow AD = 0.81 \times AC$$

$$\therefore \text{Isometric Length} = 0.81 \times \text{Actual Length}$$

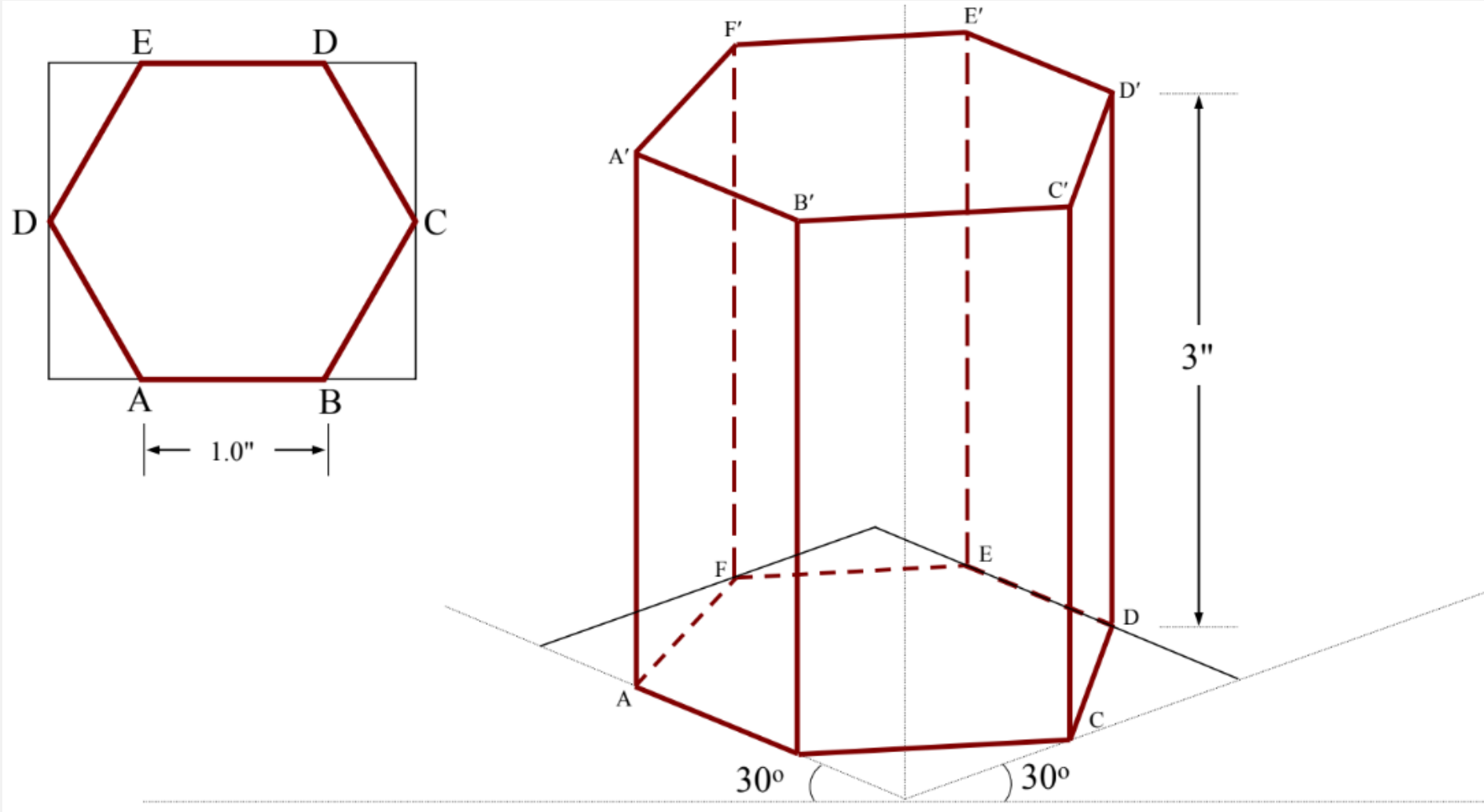
ISOMETRIC PROJECTION: DRAWING METHODS

- **Box Method:** The isometric projection of solids like cube, square and rectangular prisms is drawn directly when their edges are parallel to the three isometric axes. The isometric projection of all other types of prisms and cylinders is drawn by enclosing them in a rectangular box. This method is called Box method.
- **Co-ordinate or Offset Method:** The isometric projections of pyramids and cones are generally drawn by Co-ordinate or Offset method.

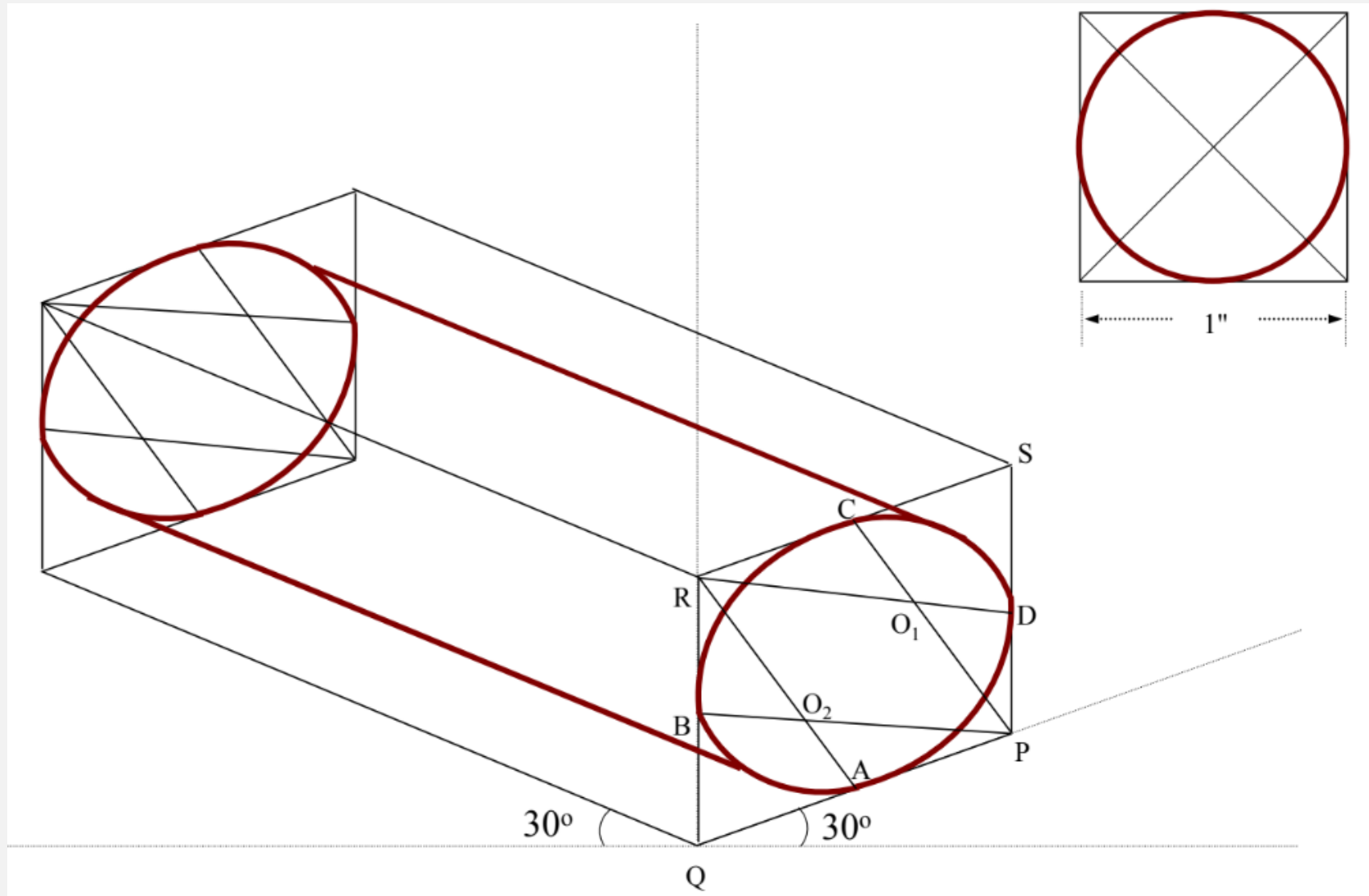
ISOMETRIC VIEW OF A REGULAR PENTAGONAL PRISM (RESTING ON ONE OF ITS FACES ON H.P)



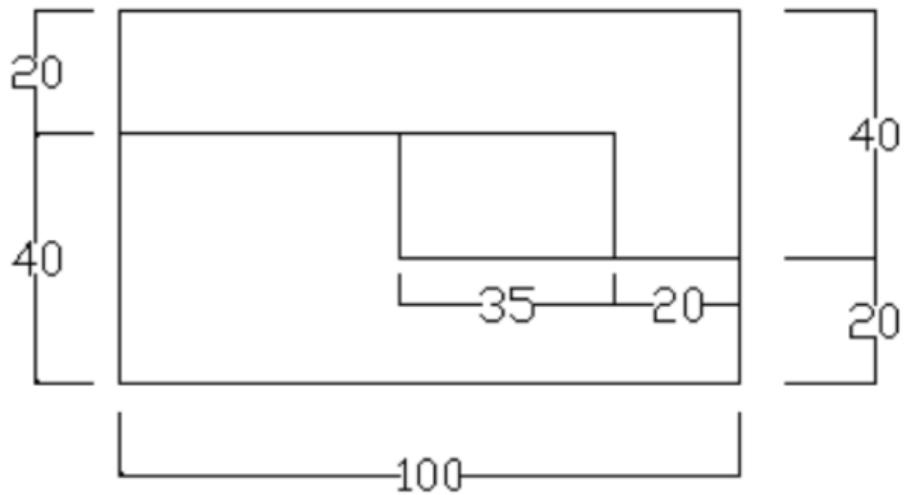
ISOMETRIC VIEW OF A REGULAR HEXAGONAL PRISM (RESTING ON ONE OF ITS FACES ON V.P)



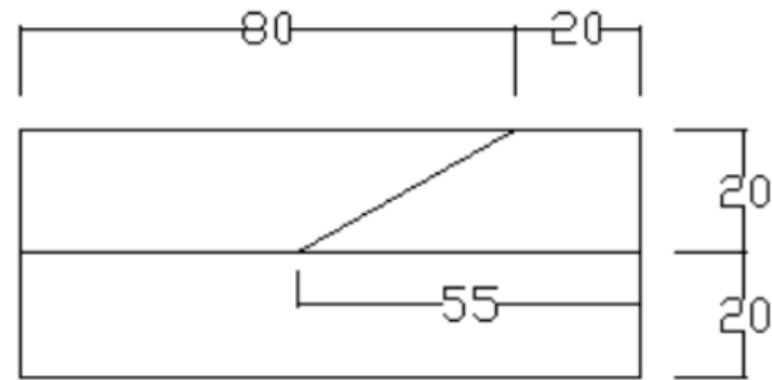
ISOMETRIC VIEW OF A CYLINDER (LYING ON H.P)



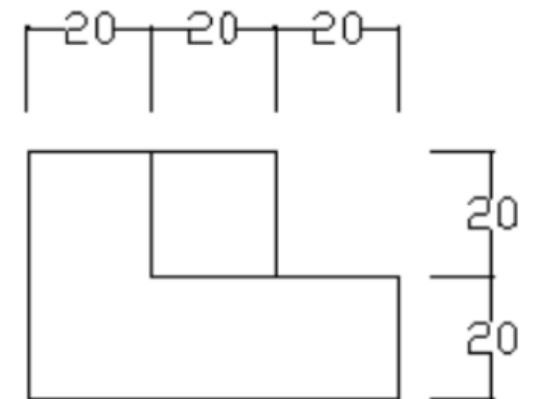
ISOMETRIC VIEW FROM THE GIVEN ORTHOGRAPHIC PROJECTION



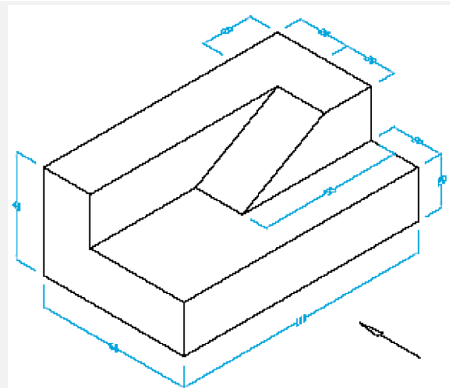
TOP VIEW



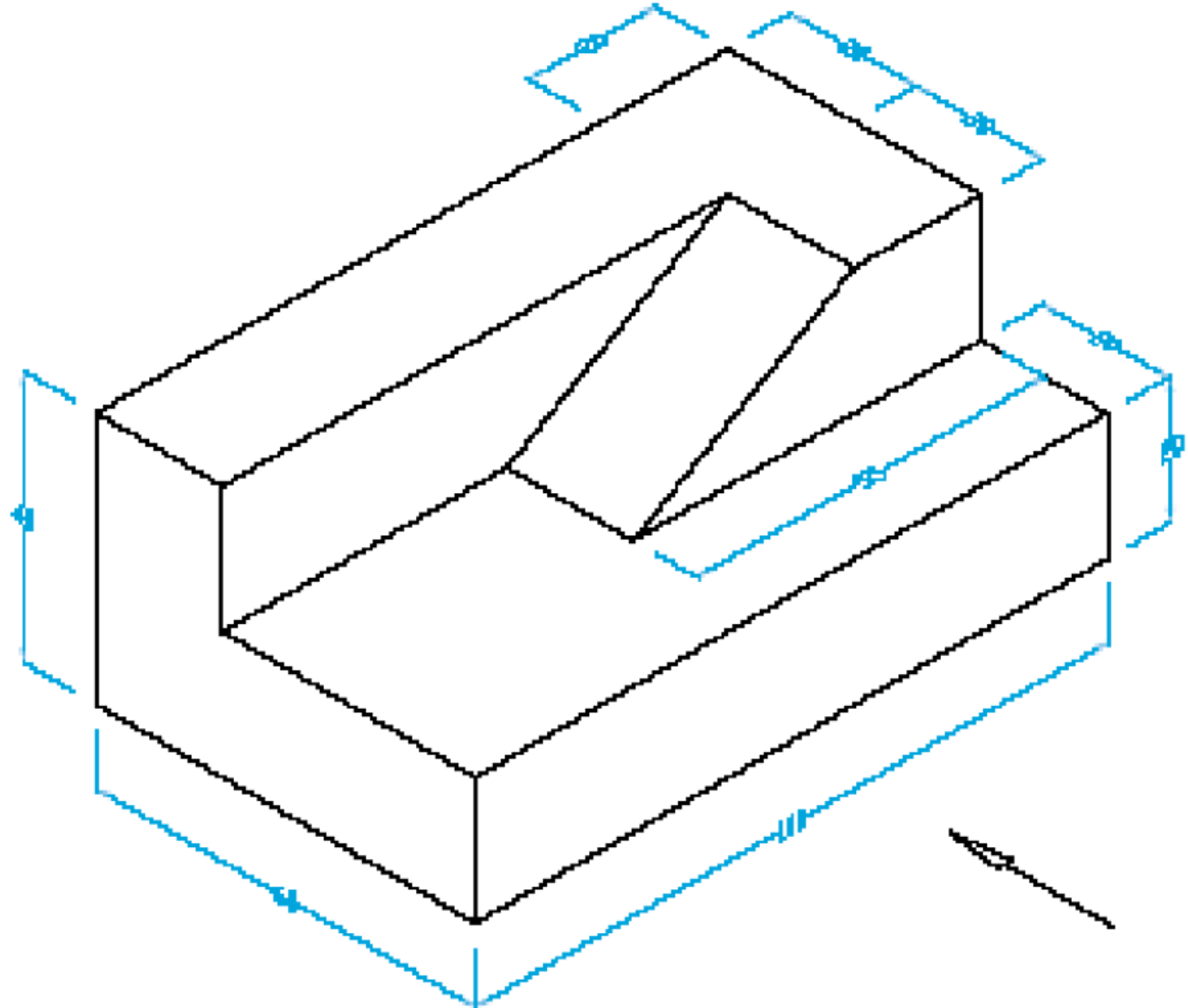
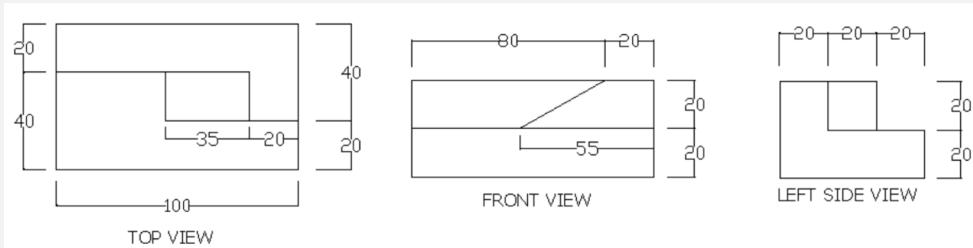
FRONT VIEW



LEFT SIDE VIEW



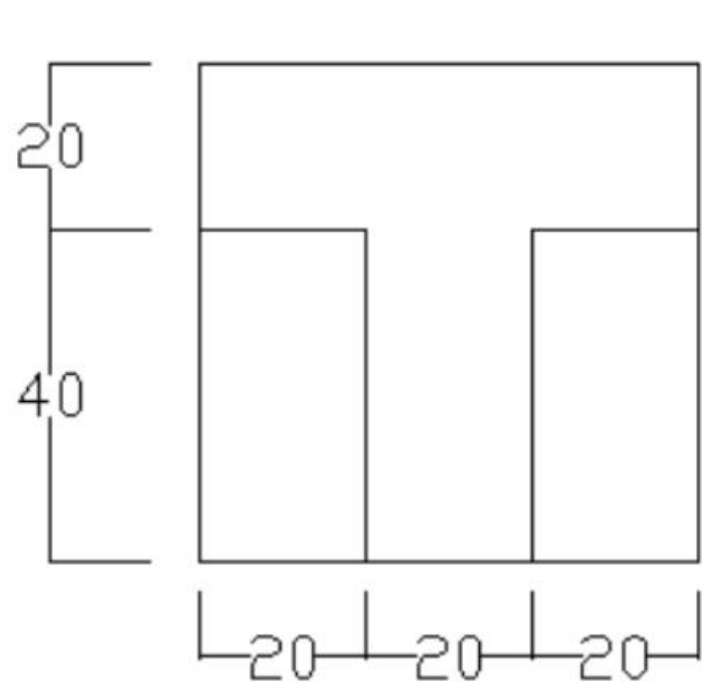
ISOMETRIC VIEW FROM THE GIVEN ORTHOGRAPHIC PROJECTION



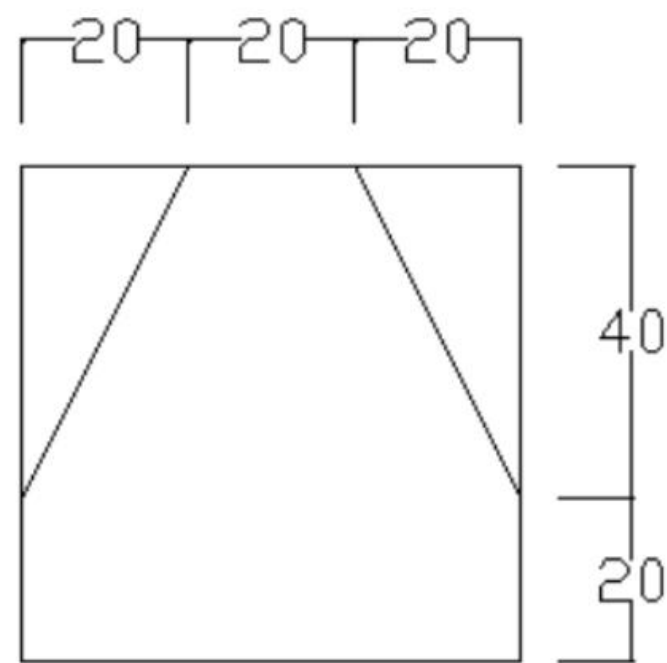
ASSIGNMENT 03

ISOMETRIC VIEW

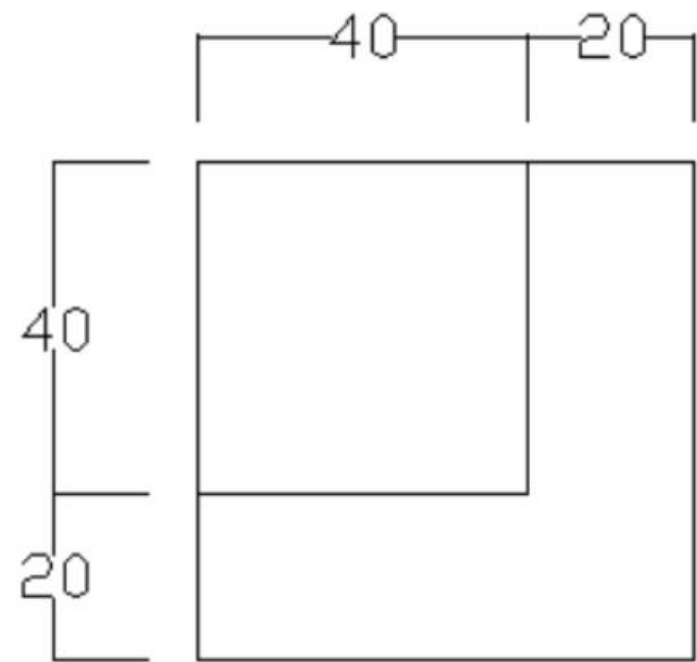
- Draw an isometric view of a cylinder (Lying on V.P)
- Draw isometric view from the given orthographic projections



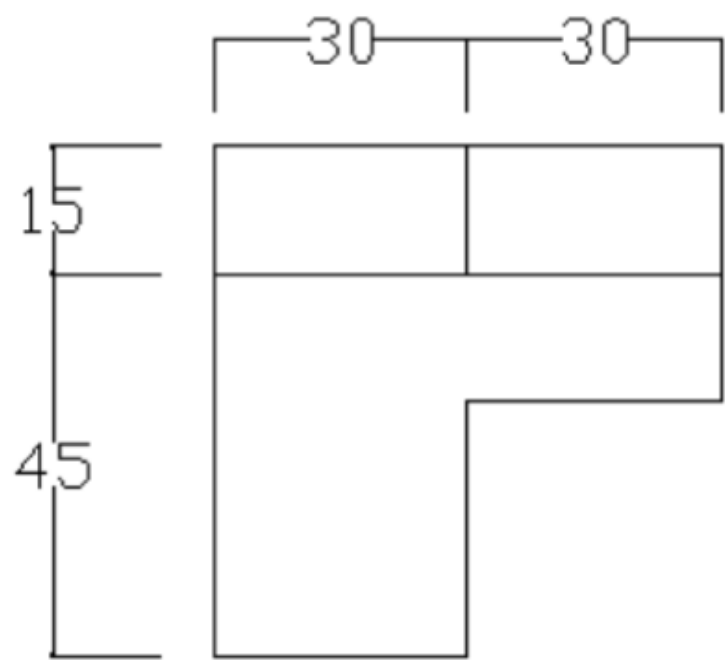
TOP VIEW



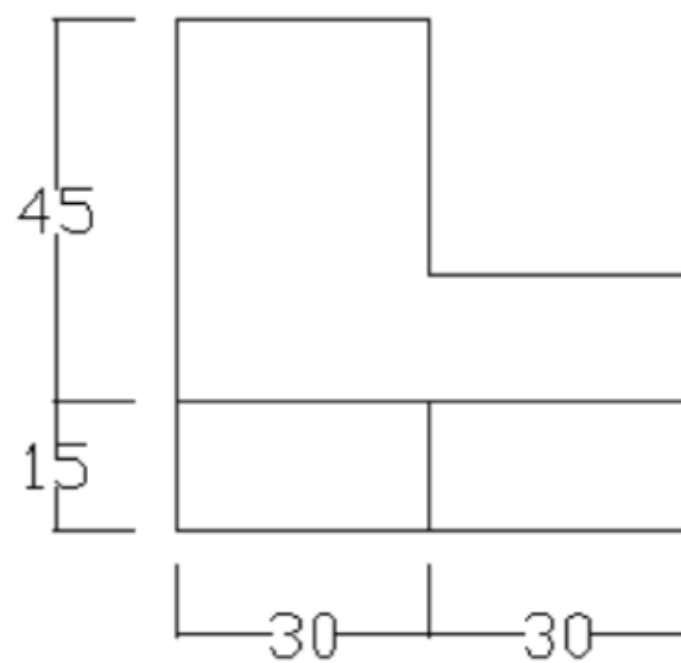
FRONT VIEW



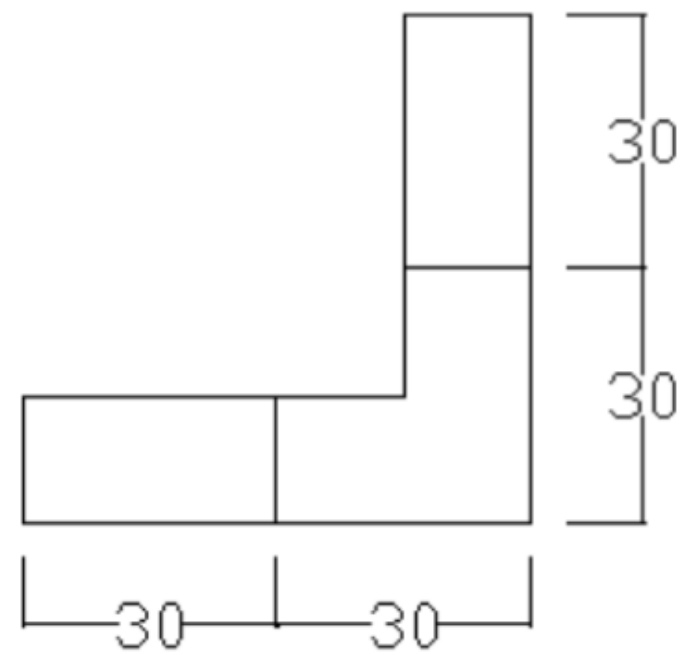
RIGHT SIDE VIEW



TOP VIEW



FRONT VIEW



RIGHT SIDE VIEW

THANK YOU